Homework 3

Combinatorics

Fall 2026

Due: Thursday, September 17, 2026 23:58
Submit: Gradescope, through Blackboard's assignment area.

Please write your homework by hand. Start each exercise on a new page, scan your work to a legible PDF, and clearly mark the pages associated with every exercise.

1. (Selecting colored balls)

Points: 10

For this exercise, an outcome records only the number of balls selected of each color. Thus balls of the same color are not distinguished from one another, and the order of selection is irrelevant.

(A) **5 points.** A box contains 5 blue balls, 7 red balls, and 14 yellow balls. How many ways are there to select a total of 15 balls?

(B) **5 points.** A box contains 5 blue balls, 7 red balls, and 20 yellow balls. How many ways are there to select a total of 15 balls?

> **Need a free hint?.** Let the b, r and y record the numbers of blue, red, and yellow balls selected. What does $b + r + y$ have to be? Are there restrictions on b, r, y?

2. (Strings avoiding specified substrings)

Points: 14

How many strings of length 15 can be formed from the 26 uppercase English letters if no letter may be repeated and none of the following strings may occur as a substring?

`CAT`, `TUG`, `GAS`, `SILK`

Here a substring consists of consecutive letters in the given order.

> **Need a free hint?.** Use inclusion–exclusion. Determine every intersection carefully. Some pairs can occur by sharing a letter: for example, a string containing both `CAT` and `TUG` must contain the block `CATUG`. Some other pairs cannot occur together because letters may not repeat.

3. (Permutations with forbidden positions)

Points: 12

The digits in the string

0123456789

are initially in their original positions. Rearrange all ten digits, using each digit exactly once.

(A) **6 points.** How many rearrangements have no digit in its original position?

(B) **6 points.** How many rearrangements satisfy all three of the following conditions?

- No digit is in its original position.
- The digits $0, 1, 2$ occupy the first three positions, in some order.
- The digits $8, 9$ occupy the last two positions, in some order.

Need a free hint?. For part (a), apply the formula for derangements. For part (b), the stated position restrictions divide the digits into the groups $\{0, 1, 2\}$, $\{3, 4, 5, 6, 7\}$, and $\{8, 9\}$. Count the allowed rearrangements within each group, then use the product rule.